

Can Variational Autoencoders disentangle Transformer Embeddings for Emotion Translation?

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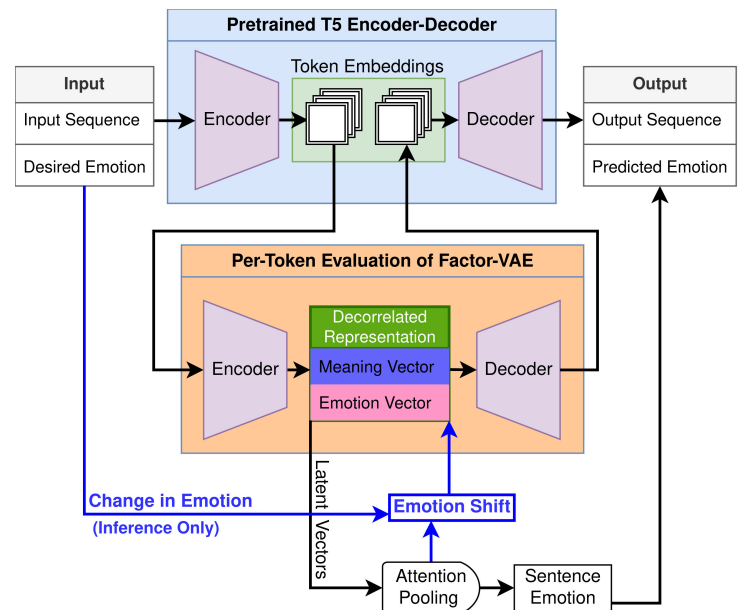
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Encoder-Decoder models based on the transformer are capable of performing a wide range of tasks by first mapping tokens to a sequence of encoder hidden states (embeddings), before remapping them autoregressively to a new sequence. Without additional inductive biases, embedding spaces suffer from a lack of interpretability, despite having a complete encoding of information. We investigate whether such embeddings can be refined and disentangled by a Variational Autoencoder (VAE). To do so, we train the VAE to reproduce the original sequence through an information bottleneck. We use a factor-VAE architecture to decorrelate the latent dimensions, which helps smooth the representation, and we include task-reserved dimensions for performing emotion-translating tasks. Particularly, we use an attention pooling component to learn a two-way mapping between some dimensions of the latents and the emotion labels, producing an emotion-translating sequence-to-sequence model. We evaluate our model using human-guided LLM labelling, which complements our reconstruction error in the translation from embedding and latents.

Architecture: we place the VAE between the encoder and decoder, forcing it to map embeddings to an interpretable latent space and then back to the original encoding. We use a VAE because it allows for new sample generation, and it can be set to enforce strong inductive biases in the latent space through a custom loss. More precisely, we set the number of latent dimensions to be half of the original embedding dimension, reserving m dimensions for encoding emotion. We then use an attention pooling block to aggregate the latent representations of the whole sequence to quantify the overall emotional tone of the sentence. During training, the Encoder-Decoder model is frozen, and the VAE first learns to compress and reconstruct the original embeddings, while the attention pooling mechanism trains to correctly identify emotion. At inference, the VAE reconstructs and classifies input sequences internally, and the attention weights are used to alter the emotion using the encoder hidden states.



We use a T5 **model** for the embeddings as it is a general purpose encoder-decoder model. We evaluate training on two **datasets**: GoEmotions and SemEval-2018 (EI-reg). The first provides 58k Reddit comments annotated with 27 fine-grained emotion categories plus Neutral, for categorical emotion detection; the latter contributes tweet-level real-valued emotion intensity scores for anger, fear, joy, and sadness, which lets us model not only which emotion is present but also how strongly it is expressed.

We test our **hypotheses** with qualitative and quantitative methods. *Does disentanglement occur?* We use DCI and MIG to evaluate it. *Does the architecture outperform prompting?* We compare our model using a human-guided classifier model: we manually label some samples as **baseline** ~100, and tweak distribution parameters of the model to obtain human-like responses. We fix the softmax probability values of the emotion classifier, treating them as ground truth and compare performance against a human baseline and prompted LLMs with different sizes. *Are all architecture components necessary?* We verify it with ablations by varying the number of attention pooling heads, the size of the latent space and the depth of the VAE.